Your airline's customer service team has been collecting chat data for years—thousands of conversations, each labeled with the user's intent and an ideal response. Now, it's time to put that data to work.

You've been tasked with fine-tuning a TinyLlama model to power the airline's next-gen AI assistant. The goal? Given a user message, the model should predict the intent (like booking a flight, checking baggage status, or requesting special assistance) and generate a helpful, human-like response. Accurate intent detection is key since it helps the system understand what the customer wants, so it can respond appropriately and trigger downstream actions when needed.

The Data

You'll work with a dataset of various travel query examples.

Column	Description
`instruction`	A user request from the Travel domain
`category`	The high-level semantic category for the intent
`intent`	The specific intent corresponding to the user instruction
`response`	An example of an expected response from the virtual assistant

Update to Python 3.10

Due to how frequently the libraries required for this project are updated, you'll need to update your environment to Python 3.10:

1. In the workbook, click on "Environment," in the top toolbar and select "Session details".
2. In the workbook language dropdown, select "Python 3.10".
3. Click "Confirm" and hit "Done" once the session is ready.

```
# First install the necessary packages
!pip install -q -q -q trl==0.16.0
!pip install -q -q -q tf-keras==2.19.0
!pip install -q -q -q peft
```

```python
# Import the required dependencies for this project
from transformers import AutoTokenizer, AutoModelForCausalLM
from trl import SFTTrainer, SFTConfig
from peft import LoraConfig

from datasets import Dataset, load_dataset
from collections import Counter, defaultdict
import random
```

```
2026-04-29 10:57:21.819993: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory:
Attempting to register factory for plugin cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1777460241.834044     381 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
E0000 00:00:1777460241.838409     381 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for
plugin cuBLAS when one has already been registered
W0000 00:00:1777460241.850000     381 computation_placer.cc:177] computation placer already registered. Please check linkage and
avoid linking the same target more than once.
W0000 00:00:1777460241.850012     381 computation_placer.cc:177] computation placer already registered. Please check linkage and
avoid linking the same target more than once.
W0000 00:00:1777460241.850014     381 computation_placer.cc:177] computation placer already registered. Please check linkage and
avoid linking the same target more than once.
W0000 00:00:1777460241.850016     381 computation_placer.cc:177] computation placer already registered. Please check linkage and
avoid linking the same target more than once.
2026-04-29 10:57:21.854478: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use
available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
```

The code below loads the travel query dataset and reduces it from ~30k to ~50 records, keeping all intent types. This speeds up fine-tuning. Run it before starting, and feel free to experiment with it later!

```python
# First load the entire dataset
ds = load_dataset('bitext/Bitext-travel-llm-chatbot-training-dataset', split="train")

# Group examples by intent
random.seed(42)
intent_groups = defaultdict(list)
for record in ds:
    intent = record["intent"]
    intent_groups[intent].append(record)

# Determine how many samples per intent
total_intents = len(intent_groups)
samples_per_intent = 100 // total_intents

# Sample from each intent
balanced_subset = []
for intent, examples in intent_groups.items():
    sampled = random.sample(examples, min(samples_per_intent, len(examples)))
    balanced_subset.extend(sampled)

total_num_of_records = 50
travel_chat_ds = Dataset.from_list(balanced_subset[:total_num_of_records])

travel_chat_ds.to_pandas().head(3)
```

```
README.md: 0.00B [00:00, ?B/s]
```

```
(…)-travel-llm-chatbot-training-dataset.csv:   0%|          | 0.00/19.7M [00:00<?, ?B/s]
```

```
Generating train split:   0%|          | 0/31658 [00:00<?, ? examples/s]
```

in…	instruction	intent	cate…	ta…
0	I'd like information about my checked baggage allowance, how can I…	check_baggage_allowance	BAGGAGE	BCI
1	i have to see the fucking checked baggage allowance can i get som…	check_baggage_allowance	BAGGAGE	BCI
2	I want to know about my checked baggage allowance, where can I fi…	check_baggage_allowance	BAGGAGE	BCI

Rows: 3 Expand

```python
# Start the project with the dataset below

print(travel_chat_ds)
```

```
Dataset({
    features: ['instruction', 'intent', 'category', 'tags', 'response'],
    num_rows: 50
})
```

```python
# 1. Import necessary libraries
from transformers import AutoModelForCausalLM, AutoTokenizer
from peft import LoraConfig
from trl import SFTTrainer, SFTConfig

# 2. Modify the dataset to include a conversation field
def merge_example(row):
    row['conversation'] = f"Query: {row['instruction']}\nIntent: {row['intent']}\nResponse: {row['response']}"
    return row

# If travel_chat_ds is a HuggingFace Dataset, use .map; else, use a list comprehension
try:
    travel_chat_ds = travel_chat_ds.map(merge_example)
except AttributeError:
    travel_chat_ds = [merge_example(row) for row in travel_chat_ds]

print(travel_chat_ds[0]['conversation'])

# 3. Load the Llama Model with local_files_only=True to avoid long downloads in notebook
model_name = "TinyLlama/TinyLlama-1.1B-Chat-v0.1"

# Fix: Check if model files are available locally, else set local_files_only=False
try:
    model = AutoModelForCausalLM.from_pretrained(model_name, local_files_only=True)
    tokenizer = AutoTokenizer.from_pretrained(model_name, local_files_only=True)
except (OSError, AttributeError):
    print(
        f"Model '{model_name}' not found locally. Downloading from HuggingFace Hub (requires internet)..."
    )
    model = AutoModelForCausalLM.from_pretrained(model_name, local_files_only=False)
    tokenizer = AutoTokenizer.from_pretrained(model_name, local_files_only=False)

tokenizer.pad_token = tokenizer.eos_token

# 4. Initialize SFTConfig
sftConfig = SFTConfig(
    max_steps=1,
    per_device_train_batch_size=4,
    gradient_accumulation_steps=1,
    learning_rate=2e-3,
    max_grad_norm=0.3,
    save_steps=10,
    dataset_text_field='conversation',
    output_dir="/tmp",
)

# 5. Initialize LoRA config
lora_config = LoraConfig(
    r=12,
    lora_alpha=32,
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
    target_modules=['q_proj', 'v_proj']
)

# 6. Initialize SFTTrainer
trainer = SFTTrainer(
    model=model,
    train_dataset=travel_chat_ds,
    peft_config=lora_config,
    args=sftConfig
)

# 7. Kickstart fine-tuning process
trainer.train()

# 8. Generate responses with fine-tuned model
inputs = tokenizer.encode("Query: I'm trying to book a flight", return_tensors="pt")
outputs = model.generate(inputs, max_new_tokens=20, max_length=20)
decoded_outputs = tokenizer.decode(outputs[0, inputs.shape[1]:], skip_special_tokens=True)
model_response = decoded_outputs
```

```
print(model_response)
```

```
Map:   0%|          | 0/50 [00:00<?, ? examples/s]
```

```
Query: I'd like information about my checked baggage allowance, how can I find it?
Intent: check_baggage_allowance
Response: To retrieve your checked baggage allowance details, please follow these instructions:

1. Visit {{WEBSITE_URL}} or launch the {{APP_NAME}} application.
2. Log in to your personal account.
3. Select the {{BOOKINGS_OPTION}} section.
4. Enter the required booking or flight information.
5. The specific baggage allowance for your trip will be displayed.

Should you require additional help, do not hesitate to contact customer support via the {{APP_NAME}} app or on {{WEBSITE_URL}}.
Model 'TinyLlama/TinyLlama-1.1B-Chat-v0.1' not found locally. Downloading from HuggingFace Hub (requires internet)...
```

```
config.json:   0%|          | 0.00/652 [00:00<?, ?B/s]
```

```
model.safetensors:   0%|          | 0.00/4.40G [00:00<?, ?B/s]
```